



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Software Engineering

Final Examination

Year 3, Semester 2 (2023)

SE3060 – Database Systems

Duration: 2 hours

November 2023

Instructions to Candidates:

- ◆ This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- ◆ This paper has 4 questions.
- ◆ Answer all questions in the booklet given.
- ◆ The total marks for the paper is 100.
- ◆ This paper contains 6 pages, including the cover page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1 – Transactions and Concurrency Control**(25 marks)**

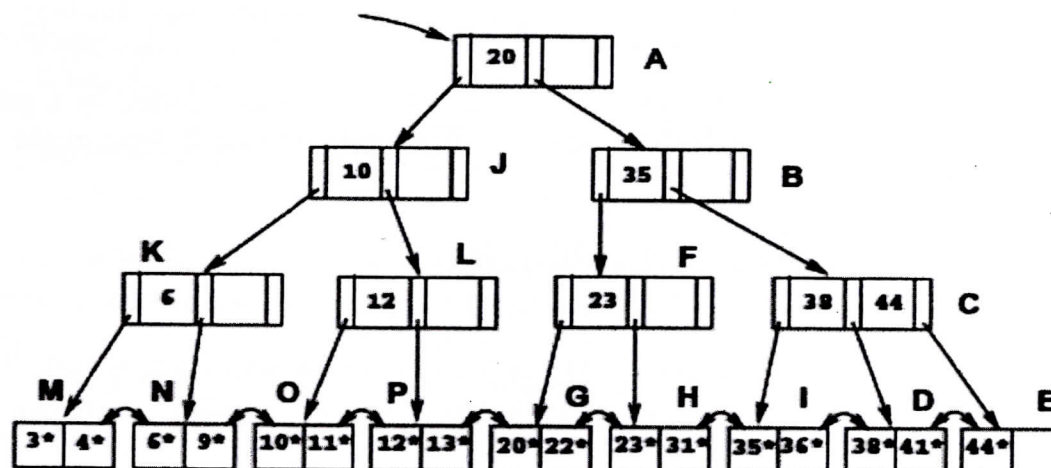
- (a) List the properties of a transaction. Briefly explain each of them. (4 marks)
- (b) Compare why serializable schedules are more important for database systems than the serial schedule. (3 marks)
- (c) “The Strict Two-Phase Locking (S2PL) protocol significantly reduces the occurrence of cascading aborts and unrecoverable schedules, enhancing concurrency control in database systems”. Do you agree with this? Provide reasonable justification for your decision. (4 marks)
- (d) Consider the following part of the transaction schedule.

T1	T2	T3	T4
S(A)			
R(A)			
		S(C)	
		R(C)	
X(C) wait			
	X(B)		
	W(B)		
		X(B) wait	
			X(D)
			W(D)
	X(D) wait		
			X(A) wait

Assume that Transaction T_i is higher priority than transaction T_{i+1} (i.e. transaction T_1 has higher priority than T_2 ; T_2 has higher priority than T_3 ; and T_3 has higher priority than T_4).

- Review the provided transaction schedule and analyse the reasons for the delay in completing these transactions. Assume that the strict two-phase locking protocol has been used to enable concurrency. (3 marks)
- Draw a Wait-For-Graph for the given schedule and check for deadlock. (2 marks)
- Draw the schedule again considering deadlock prevention algorithm: Wound-wait approach. (4 marks)

(e) Consider the tree shown below.



Describe the steps involved in executing each of the following operations according to the *Simple Tree Locking algorithm*, in terms of the order in which nodes are locked, unlocked, read and written. Be specific about what type of locks are obtained and released.

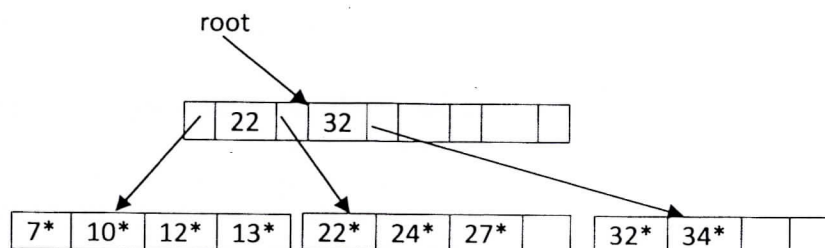
- Search 23.
- Delete 43.
- Insert 44.

(5 marks)

Question 2 – Indexing Techniques

(25 marks)

- Briefly explain three file organization techniques used in database systems to store and manage data efficiently.
(4 marks)
- Given that indexes can speed up queries, why don't we just create as many indexes as we like?
(3 marks)
- How is a clustered B+ tree index different from an un-clustered one?
(3 marks)
- Consider the following B+ tree of order 2:



Illustrate the B+ tree after deleting 32. Consider left sibling for redistribution of data entries.

(5 marks)

- (e) Construct an Extendible Hashed File for the following marks received by a group of students in a course. The hashing function will consider the last 2 digits of the binary representation.

marks	51	58	62	66	50	54
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(5 marks)

- (f) Provide three reasons to justify why most commercial databases support B+ tree indexes.

(5 marks)

Question 3 – Query Optimization

(25 marks)

- (a) What is the justification for using I/O costs as the main measure to compare different algorithms for evaluating relational operators?
- (2 marks)
- (b) Briefly explain why it is often advantageous to do selections before joins in a query plan. How do early projections help during the query execution?
- (3 marks)

- (c) Consider the following relation:

Emp (*eid*: integer, *ename*: string, *age*: integer, *salary*: float)

On this table, there is a *clustered* B+ tree index on *eid* and an *unclustered* B+ tree index on *age*.

- i. The data records are stored in a heap file. In what order are the data records of Emp table stored and why?
- (2 marks)

- ii. Consider the following query:

```
Select ename, salary
From emp
Where age>20;
```

If 95% of tuples satisfy the selection condition, what would be the best access path for processing this query? Justify your answer.

(5 marks)

iii. Consider the following query:

```
Select age, avg(salary)
From emp
Group by age;
```

Would a clustered index on <age, salary> be useful? Explain briefly.

(6 marks)

- (d) The relation Emp (eid: integer, ename:string, age: integer, salary: float) contains 4,000 pages. There are 10 buffer pages. The attribute *eid* takes up 10 bytes, *ename* 20 bytes, *age* 2 bytes and *salary* 8 bytes. A clustered B+ tree on <age, salary> is the **only index** available. In estimating the cost of query plans, ignore the cost of writing the final result.

Consider the following query:

```
Select age, avg(salary)
From emp
Where salary > 40000
Group by age;
```

Assume that 10% of the tuples satisfy the selection condition. Describe the best plan and show an estimation of its cost.

(7 marks)

Question 4 – Big data

(25 marks)

- (a) Discuss two real world examples where NoSQL databases can be applied.
(5 marks)
- (b) Consider the following structure of books collection and write answers for the following questions.

```
{
  book_id : "B001" ,
  title : "MongoDB: The Definitive Guide" ,
  author : "Kristina Chodorow" ,
  pages : 432 ,
  language : "English" ,
  year : 2013 ,
  publisher : {
    name : "O'Reilly Media" ,
    founded : 1980 ,
    location : "CA"
  }
}
```

- i. Write a MongoDB query to display the fields *book_id*, *title*, *author* and *year* for all the documents in the books collection.

(2 marks)

- ii. Write a MongoDB query to display all the books which are published in the English language.
(2 marks)
- iii. Write a MongoDB query to find the books that have pages, more than 100 but less than 500.
(2 marks)
- iv. Write a MongoDB update statement to change the publication year of the book with 'book_id' 'B001' from 2013 to 2015.
(4 marks)
- (c) Compare single field indexes and multikey indexes in MongoDB.
(5 marks)
- (d) Briefly, explain four modules of the Hadoop framework.
(5 marks)

- End of the Question Paper -